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Aim:- To write a LEX program to count the number of blank spaces, words & lines in given file.

Algorithm:-

1. Start
2. Initialize counters for words, spaces & lines
3. Define patterns for words, spaces & lines
4. Read file using LEX.
5. Increment counters accordingly.
6. Print results
7. Stop.

Program:-

```
%{
#include <stdio.h>
int words = 0, spaces = 0, lines = 0;
%}

%%
[ \t ]+ { spaces++; }

\n { lines++; }

[a-z A-Z]+
{
words++;
}
```

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input Hello GIT
This is Kanchan

Output

Words: 5
lines : 2
Spaces: 3


```
$
}
%%

int main()
{
    yylex();
    printf("Words: %d\n", words);
    printf("Spaces: %d\n", spaces);
    printf("Lines: %d\n", lines);
    return 0;
}
```

Result we have successfully completed code for count the lines, spaces & words in code.

Viva Questions :-

Q1 How do you define a word in your LEX program?
Ans A word is defined as using regular expression that matches a sequences of alphabatic characters. For example, the patten [a-zA-Z]+ represents one or more uppercase or lowercase letters. Whenever this pattern is matched, the word count is incremented.

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Q2 How do you count blank lines?

Ans Blank lines are identified by detecting newline characters that do not contain any other visible characters. A suitable LEX pattern can match an empty line or a line containing only spaces/tabs. Whenever such a pattern is found, the blank-line counter is incremented.

Q3 What is the difference between counting words & counting lines?

Ans Words are counted by matching sequences of characters that form individual words, such as [a-zA-Z]+, lines are counted by detecting the newline character \n. Therefore, a single line may contain multiple words, but it is counted as only one line.

Q4 How do you handle multiple consecutive spaces?

Ans Multiple consecutive spaces can be handled using a pattern such as [\t]+, which matches one or more spaces or tabs together. These spaces are generally ignored & are not considered words. This prevents extra spaces from affecting the word count.

Q5 What is the purpose of the yywrap() function in LEX program?

Ans yywrap() function is called when the lexical analyzer reaches the end of the input files. If it returns 1 LEX understand there is no more input to process. If 0, then Lexer expects more inputs.

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